

Track 2D Predicted Mean h0 Surface Diagnostic

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

Predicted-mean versus measured- h_0 diagnostic over 39 selected candidates from the Track 2D per-curve matrix.

Decision

- The next intervention should target predicted mean / offset-surface behavior, not measured h_0 magnitude alone.
- Median selected-candidate mean bias is -0.0001 deg and median mean absolute offset error is 0.0031 deg.
- Median predicted-mean versus measured- h_0 slope is 0.9769 ; 1 selected candidates deviate from unit slope by at least 0.50 .
- 1 selected candidates have absolute mean bias at least 0.005 deg .

Candidate Selection

Candidate	Reason
DT19_Bw	surface_leader_Bw
ELM19_Bw	largest_mean_offset
ELM19_Fw	strong_h0_error_overlap
GBM19_Fw	strong_h0_error_overlap
LGBM19_Fw	strong_h0_error_overlap
MLP19_Bw	largest_mean_offset
MLP19_Fw	largest_mean_offset
XGBM19_Bw	largest_mean_offset
XGBM19_Fw	strong_h0_error_overlap
feedforward_global	global_high_offset
gru_sequence_global	global_high_offset
harmonic_regression_global	global_high_offset;largest_mean_offset
paper_retuned_best_Fw	track2d_top_rank
periodic_gru_sequence_global	surface_leader_global
periodic_temporal_convolution_global	global_high_offset
rcim_original_DT19_Fw	track2d_top_rank
rcim_original_ERT19_Fw	track2d_top_rank
rcim_original_GBM19_Fw	track2d_top_rank

Candidate	Reason
rcim_original_LGBM19_Fw	track2d_top_rank
rcim_original_MLP19_Fw	largest_mean_offset
rcim_original_RF19_Fw	track2d_top_rank
rcim_retuned_DT19_Fw	track2d_top_rank
rcim_retuned_ELM19_Bw	largest_mean_offset;strong_h0_error_overlap
rcim_retuned_ERT19_Fw	track2d_top_rank
rcim_retuned_GBM19_Fw	surface_leader_Fw;track2d_top_rank
rcim_retuned_HGBM19_Bw	strong_h0_error_overlap
rcim_retuned_LGBM19_Bw	strong_h0_error_overlap
rcim_retuned_MLP19_Bw	largest_mean_offset
rcim_retuned_MLP19_Fw	largest_mean_offset
rcim_retuned_RF19_Fw	track2d_top_rank
rcim_retuned_XGBM19_Bw	largest_mean_offset;strong_h0_error_overlap
residual_harmonic_gru_sequence_dense240_global	global_high_offset
residual_harmonic_gru_sequence_dense360_global	global_high_offset
residual_harmonic_lstm_sequence_dense240_global	global_high_offset
residual_harmonic_lstm_sequence_dense360_global	global_high_offset
residual_harmonic_lstm_sequence_sparse_rcim_Fw	strong_h0_error_overlap
residual_harmonic_mlp_global	global_high_offset
temporal_convolution_Bw	strong_h0_error_overlap
temporal_convolution_global	global_high_offset

Candidate Mean-Surface Summary

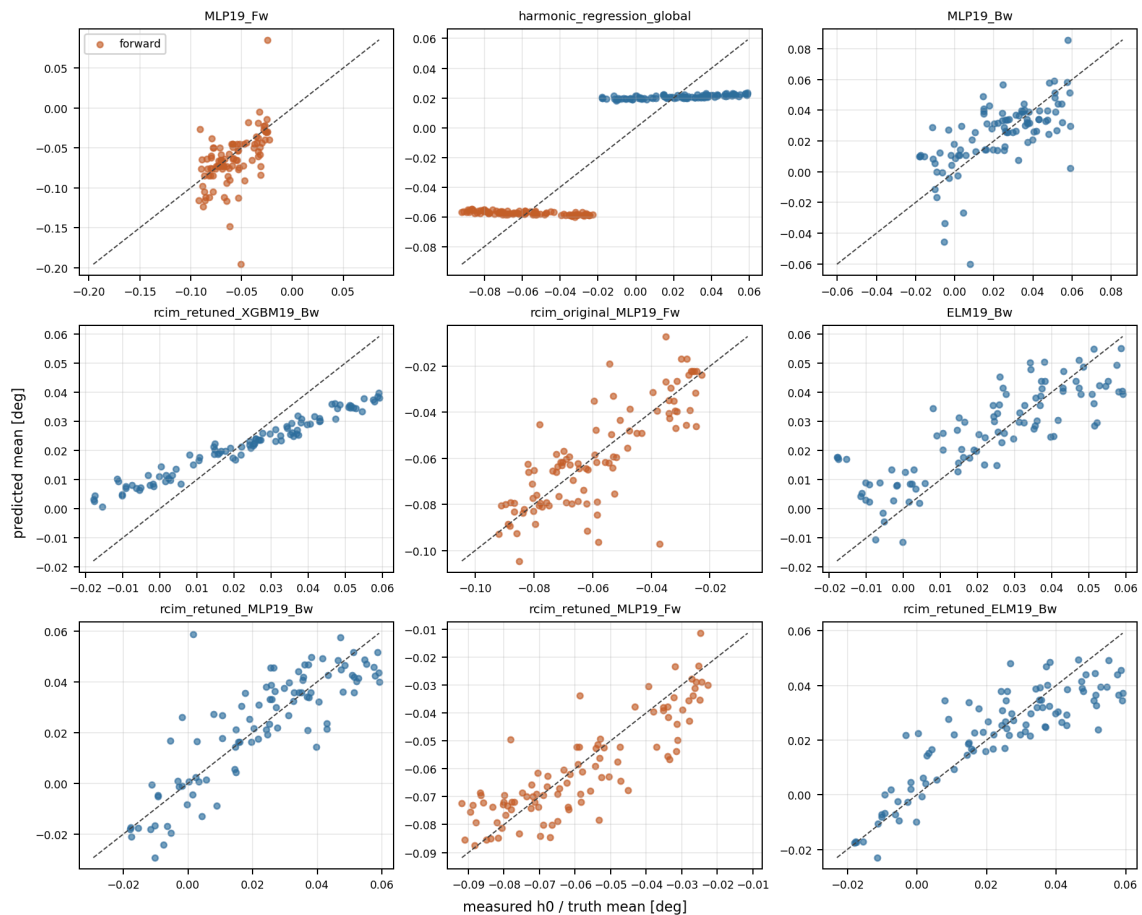
Candidate	Surface	Bias	Mean AE	P90 AE	Corr	Slope	Intercept
MLP19_Fw	Fw	-0.0060	0.0185	0.0413	0.5330	0.8766	-0.0132
harmonic_regression_global	global	-0.0005	0.0180	0.0327	0.8892	0.7598	-0.0047
MLP19_Bw	Bw	0.0013	0.0137	0.0283	0.6407	0.6408	0.0096
rcim_retuned_XGBM19_Bw	Bw	-0.0013	0.0101	0.0196	0.9826	0.4648	0.0110
rcim_original_MLP19_Fw	Fw	-0.0006	0.0098	0.0207	0.8007	0.8802	-0.0076
ELM19_Bw	Bw	0.0033	0.0097	0.0187	0.8343	0.6028	0.0124
rcim_retuned_MLP19_Bw	Bw	-0.0000	0.0090	0.0190	0.8458	0.8674	0.0030
rcim_retuned_MLP19_Fw	Fw	-0.0015	0.0083	0.0177	0.8556	0.7450	-0.0165
rcim_retuned_ELM19_Bw	Bw	-0.0010	0.0081	0.0188	0.8713	0.7220	0.0054
XGBM19_Bw	Bw	0.0030	0.0077	0.0171	0.9240	0.5587	0.0131

Direction Split

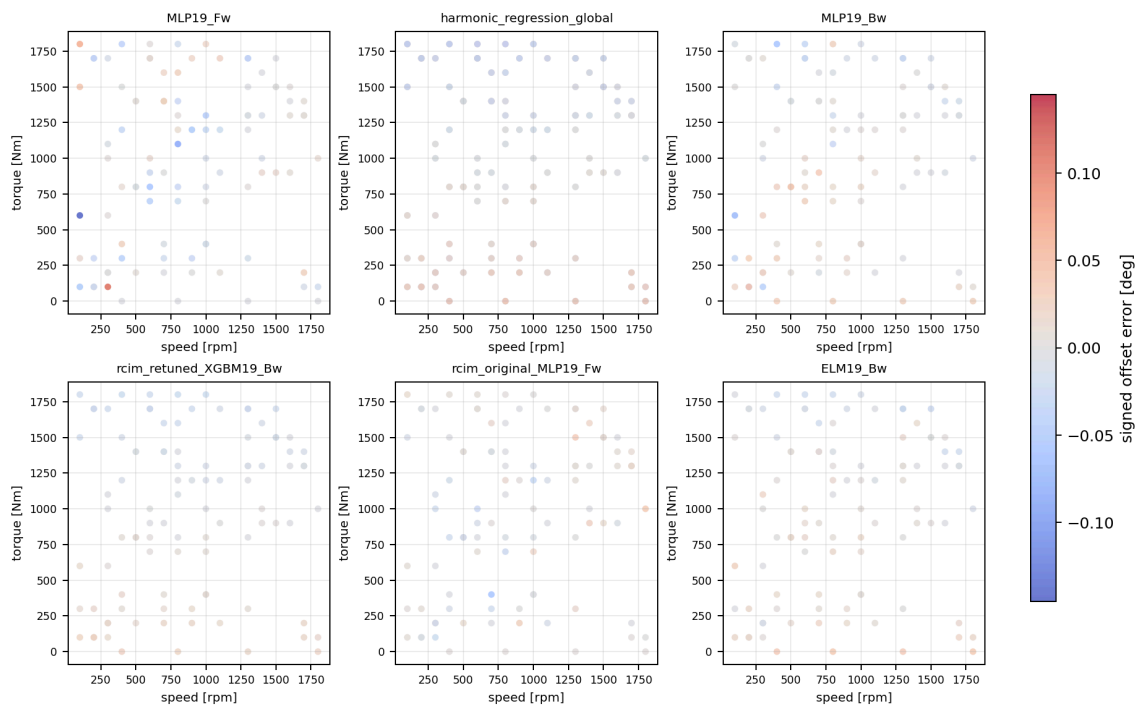
Candidate	Direction	Bias	Mean AE	Corr	Slope
ELM19_Bw	backward	0.0033	0.0097	0.8343	0.6028
MLP19_Bw	backward	0.0013	0.0137	0.6407	0.6408
MLP19_Fw	forward	-0.0060	0.0185	0.5330	0.8766
XGBM19_Bw	backward	0.0030	0.0077	0.9240	0.5587

Candidate	Direction	Bias	Mean AE	Corr	Slope
harmonic_regression_global	backward	-0.0021	0.0178	0.7416	0.0410
harmonic_regression_global	forward	0.0011	0.0182	-0.7018	-0.0411
rcim_original_MLP19_Fw	forward	-0.0006	0.0098	0.8007	0.8802
rcim_retuned_ELM19_Bw	backward	-0.0010	0.0081	0.8713	0.7220
rcim_retuned_MLP19_Bw	backward	-0.0000	0.0090	0.8458	0.8674
rcim_retuned_MLP19_Fw	forward	-0.0015	0.0083	0.8556	0.7450
rcim_retuned_XGBM19_Bw	backward	-0.0013	0.0101	0.9826	0.4648

Visual Diagnostics



Predicted mean versus measured h0



Signed offset error by speed and torque

Interpretation

The useful signal is model-side mean-surface behavior: candidates with large offset error generally show candidate-specific bias, compressed or shifted predicted-mean surfaces, or direction-dependent behavior. This supports planning an offset/mean head or calibration branch that is evaluated per `Fw`, `Bw`, and `global` surface, while keeping centered-shape metrics separate.

Machine-Readable Artifacts

- `output/validation_checks/track2d_predicted_mean_h0_surface_diagnostic/2026-06-10-13-00-59__track2d_predicted_mean_h0_surface_diagnostic/track2d_predicted_mean_h0_candidate_summary.csv`
- `output/validation_checks/track2d_predicted_mean_h0_surface_diagnostic/2026-06-10-13-00-59__track2d_predicted_mean_h0_surface_diagnostic/track2d_predicted_mean_h0_direction_summary.csv`
- `output/validation_checks/track2d_predicted_mean_h0_surface_diagnostic/2026-06-10-13-00-59__track2d_predicted_mean_h0_surface_diagnostic/track2d_predicted_mean_h0_candidate_selection.csv`
- `output/validation_checks/track2d_predicted_mean_h0_surface_diagnostic/2026-06-10-13-00-59__track2d_predicted_mean_h0_surface_diagnostic/track2d_predicted_mean_h0_selected_rows.csv`
- `output/validation_checks/track2d_predicted_mean_h0_surface_diagnostic/2026-06-10-13-00-59__track2d_predicted_mean_h0_surface_diagnostic/track2d_predicted_mean_h0_surface_diagnostic_summary.yaml`

Reproduction

```
powershell conda run -n pinns_env python -B  
scripts/reports/analysis/build_track2d_predicted_mean_h0_surface_diagnostic.py
```